Research Article

On the performance of Power-Splitting SWIPT AF Relay-Assisted NOMA System with Imperfect SIC

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Abstract. *A two-user downlink non-orthogonal multiple access (NOMA) system assisted by an energy-harvesting relay is investigated to characterize reliability and throughput under realistic wireless impairments. In which, the relay employs a power-splitting simultaneous wireless information and power transfer (PS-SWIPT) architecture to harvest energy from the received signal and forwards the superimposed NOMA waveform using amplify-and-forward (AF) relaying over block Rayleigh fading channels with distance-dependent path loss, while imperfect successive interference cancellation (SIC) is captured at the user side. In this work, reliability is quantified using outage probability (OP), which represents the likelihood that the instantaneous received SINR falls below a target threshold. Throughput performance is characterized by ergodic capacity (EC), which represents the average achievable transmission rate over fading realizations. Closed-form expressions are developed for the OP of both users, and EC expressions are obtained using low-complexity numerical approximation to handle analytically intractable integrals. Numerical results confirm that the derived analysis tightly matches Monte-Carlo simulations and reveal key design tradeoffs: outage probability decays with the transmit signal-to-noise ratio (SNR), power allocation induces a reliability imbalance between users, and the PS factor exhibits an optimal intermediate region that simultaneously minimizes outage and maximizes EC. These outcomes provide practical guidelines for configuring power allocation and PS in wireless-powered cooperative NOMA networks.*

Keywords

Non-orthogonal multiple access (NOMA); simultaneous wireless information and power transfer; imperfect successive interference cancellation (SIC); amplify-and-forward (AF) relaying protocol; performance analysis

1. Introduction

Non-orthogonal multiple access (NOMA) has remained a leading multiple-access strategy for beyond-fifth-generation (B5G) and sixth-generation (6G) systems because it multiplexes users in the power domain and then relies on successive interference cancellation (SIC) to separate superimposed signals at the receivers [1]. Accordingly, cooperative non-orthogonal multiple access (C-NOMA) has attracted sustained interest be-

cause relaying is able to extend coverage [2,3] while also stabilizing quality-of-service (QoS) for cell-edge users [1]. Nevertheless, practical implementations of SIC are often imperfect, and residual interference therefore persists and can dominate reliability and throughput, especially in multiple system components [4].

In parallel with spectral-efficiency targets, energy sustainability has become equally decisive in Internet-of-Things (IoT) deployments, where battery replacement is costly and sometimes infeasible [5]. Wireless power transfer therefore becomes a useful mechanism because it can replenish energy-constrained nodes without physical intervention, and it can be embedded into relay-assisted architectures to support continuous operation [6]. Moreover, simultaneous wireless information and power transfer (SWIPT) becomes particularly attractive because it enables a receiver to extract information and harvest energy from the same radio-frequency signal, thereby coupling the information path and the power supply path by design [7, 8]. In addition, mobile edge computing has been increasingly combined with NOMA and SWIPT so that communication, computation, and energy constraints are handled within a unified optimization loop [9].

1.1. Related Works

Recent research has examined SWIPT-enabled cooperative NOMA from several complementary perspectives. In [10], SWIPT-based power splitting is integrated into coordinated NOMA and is analyzed to quantify outage and rate behavior under fading, which in turn clarifies how energy transfer parameters reshape spectral utilization. In [11], two relay selection schemes are developed with decode-and-forward (DF) relaying in a SWIPT-NOMA network, and an optimized power-splitting factor is developed to improve the throughput and energy efficiency under both perfect SIC and imperfect SIC. In [12], a two-tier cooperative NOMA network with SWIPT is formulated under explicit outage probability (OP) constraints, and a successive convex approximation method is used to obtain feasible resource-allocation policies.

Nonlinear harvesting and practical impairment modeling have also been emphasized. In [13], the near user operates as a battery-assisted non-linear energy-harvesting (EH) relay, and closed-form outage and throughput expressions are derived under imperfect SIC, which demonstrates that non-linear harvesting can reduce diversity and that a direct link can mitigate this loss. In [14], a SWIPT-assisted cooperative NOMA network is analyzed under a non-linear EH model with multiple interferers, and closed-form expressions are derived for OP and throughput under imperfect SIC, thereby quantifying how interference simultaneously affects harvested energy and decoding reliability. In [15], full-duplex (FD) cooperative SWIPT-NOMA using time-switching relaying (TSR) is investigated and explicit outage-performance expressions are obtained, which indicates that switching ratios and EH efficiency strongly shape reliability trends.

At the network-architecture level, relay-only designs have been extended to hybrid and intelligent-surface-assisted solutions. In [16], hybrid active relays coexisting with reconfigurable intelligent surfaces (RISs) are investigated in NOMA networks, and a closed-form outage analysis is developed alongside node selection under QoS constraints. In [17], a coexisting passive RIS and active relay assisted NOMA framework is introduced, and joint optimization is performed to enhance sum rate and max-min fairness through coordinated power allocation and passive beamforming.

Internet-of-things (IoT)-centric cooperative NOMA studies have increasingly adopted explicit powering entities and interference realism. In [18], an IoT access point is powered by a dedicated beacon and is evaluated under co-channel interference, where exact and asymptotic outage expressions and an ergodic-data metric are derived and then used for low-complexity parameter optimization and learning-based prediction. In [19], SWIPT-aided cooperative relaying NOMA is studied for consumer IoT under compound fading, and the analysis shows that optimizing the power-splitting and power-allocation factors is essential for maximizing ergodic sum rate.

At the physical-layer modeling level, SIC realism has been revisited under Rayleigh fading. In [4], the dependence between post-SIC noise and fading is explicitly modeled, and exact OP together with an ergodic-capacity (EC) expression is derived, which shows that common residual-interference abstractions can misestimate performance under practical power allocation. In [20], end-to-end learned NOMA signaling is developed for Rayleigh fading with quantized channel state information (CSI), and the results illustrate how limited feedback changes interference-aware constellation design.

Complementary results reinforce the continuing emphasis on tractable reliability analysis under non-ideal CSI/SIC. In [21], OP for simultaneously transmitting and reflecting RIS (STAR-RIS)-aided NOMA is derived under imperfect CSI with asymptotic analysis to characterize high signal-to-noise ratio behavior. In [22], a RIS-aided device-to-device (D2D) NOMA framework is evaluated and outage expressions are derived under imperfect SIC, which highlights the operational impact of residual interference. In [23], closed-form outage expressions are derived for partial-NOMA backscatter D2D links over Rayleigh fading, thereby clarifying the role of power allocation and bandwidth partition-

Tab. 1: Comparison with related SWIPT-NOMA relay systems.

Ref.	Relay	SWIPT	Imp. SIC	OP	EC	Key Feature
[11]	DF	Yes	Yes	Yes	Yes	Relay selection
[13]	FD	Yes	Yes	Yes	Yes	Nonlinear EH
[14]	Coop.	Yes	Yes	Yes	Partial	Interference-aware
[15]	FD	Yes	Yes	Yes	Partial	TSR protocol
Our	AF	PS-SWIPT	Yes	Yes	Yes	AF noise + PS-SWIPT

ing. In [24], RIS-assisted multi-antenna reception is analyzed over Rayleigh fading and closed-form outage expressions are developed using a Gamma approximation, which provides a reusable analytical pattern for reliability modeling. In [25], EH-enabled long-range links are analyzed under hardware impairments, which confirms that non-ideal transceiver effects can dominate performance even when EH is feasible.

Overall, all of discussed studies (spanning form 2023 to 2025) have significantly advanced the understanding of SWIPT-enabled cooperative NOMA systems under various relay architectures, harvesting models, and interference conditions. However, AF relay-assisted SWIPT-NOMA systems remain less explored than their DF counterparts, particularly when OP and ergodic capacity (EC) are jointly analyzed under imperfect SIC. This observation motivates the development of the analytical framework presented in this work.

1.2. Research Gap, Motivation, and Contributions

As summarized in Table 1, existing SWIPT-NOMA studies mainly focus on DF relaying, relay selection, nonlinear EH modeling, or system-level resource optimization. Although these studies provide important insights, the combined impacts of amplify-and-forward (AF) noise amplification, power-splitting (PS)-SWIPT operation, and imperfect SIC (Imp. SIC) have not been jointly characterized in a unified analytical framework. Moreover, most existing works emphasize either reliability metrics or throughput metrics, whereas a simultaneous OP-and-EC characterization under block Rayleigh fading remains limited. In many contemporary SWIPT-NOMA designs, the relay is based on DF operation or relay selection, and the AF amplification coupling is therefore not the primary focus [26]. In several system-level studies, the main objective is latency or resource optimization, and closed-form end-to-end reliability/throughput characterization under AF coupling is not central [6]. Furthermore, modern SIC-aware analyses indicate that the post-SIC statistics can become dependent and thus require careful signal-to-interference-plus-noise ratio (SINR) modeling before drawing design conclusions [4].

Unlike most existing SWIPT-NOMA studies that primarily consider DF relaying, relay selection, or system-level resource optimization, this work focuses on the joint analytical characterization of power-splitting SWIPT, AF relaying, and imperfect SIC within a unified framework. The main technical challenge originates from the coupling between harvested energy, relay amplification, forwarded relay noise, and residual interference caused by imperfect SIC. Existing studies typically investigate only a subset of these effects, whereas the proposed model captures their simultaneous impact on both outage probability and ergodic capacity. Consequently, this work provides a tractable reliability-throughput characterization for AF-SWIPT-NOMA systems and offers practical insights into the joint design of power allocation and power-splitting parameters under realistic SIC conditions. The main contributions are summarized as follows:

- **AF SWIPT-NOMA model under Rayleigh fading:** A PS-based SWIPT relay is modeled for forwarding a superimposed two-user NOMA signal using AF. Unlike existing DF-oriented analyses, the proposed model explicitly captures the coupling among harvested energy, relay transmit power, AF-amplified noise, and imperfect SIC.
- **Outage analysis with imperfect SIC:** Tractable OP expressions are derived for both users under imperfect SIC, which enables identification of regimes where residual interference or EH randomness becomes the performance bottleneck.
- **Ergodic capacity characterization:** EC expressions are obtained via Gaussian-Chebyshev quadrature so that numerical evaluation remains fast and stable when exact closed forms become analytically intractable.

Herein, the OP serves as the reliability metric, whereas EC serves as the throughput metric throughout this paper.

2. System Model

We consider a cooperative SWIPT-enabled NOMA network consisting of one source S, one energy-harvesting AF relay R, and two users U_1 and U_2. The relay employs a PS architecture to simultaneously harvest energy and process information.

All wireless links experience block Rayleigh fading: the channel coefficient remains constant within one transmission block and changes independently from block to block. Let h_{XY} denote the complex channel coefficient from node X to node Y, where $X \in \{S, R\}$ and $Y \in \{R, U_1, U_2\}$. Define the channel power gain

$g_{XY} \triangleq |h_{XY}|^2$. Under Rayleigh fading, the channel power gain g_{XY} follows an exponential distribution with mean Ω_{XY}, which is a standard result widely adopted in wireless communication system analysis [27, 28]. Accordingly, the CDF and PDF of g_{XY} can be written as

$$F_{g_{XY}}(x) = 1 - \exp\left(-\frac{x}{\Omega_{XY}}\right), \qquad x \geq 0, \tag{1}$$

$$f_{g_{XY}}(x) = \frac{1}{\Omega_{XY}} \exp\left(-\frac{x}{\Omega_{XY}}\right), \qquad x \geq 0. \tag{2}$$

To account for large-scale path-loss, we model the mean channel gain as $\Omega_{XY} = d_{XY}^{-\zeta}$, where d_{XY} is the distance between nodes X and Y, and ζ is the path-loss exponent. This distance-dependent large-scale fading model is commonly employed in cooperative wireless network analysis and SWIPT-enabled communication systems [27, 28].

The considered transmission protocol is executed over two consecutive phases. In the first phase, the source broadcasts a superimposed NOMA signal to the energy-harvesting relay. The relay simultaneously performs information processing and energy harvesting through the PS-SWIPT architecture. In the second phase, the relay uses the harvested energy to amplify and forward the received NOMA signal toward both users. Finally, U_1 directly decodes its intended signal, whereas U_2 first decodes the signal of U_1, performs SIC, and then decodes its own signal.

2.1. Transmit Signal Model

The source transmits a superimposed NOMA signal

$$x_S = \sqrt{P_S}\left(\sqrt{\alpha_1}s_1 + \sqrt{\alpha_2}s_2\right), \tag{3}$$

where P_S is the source transmit power, s_i ($i \in \{1, 2\}$) is the information symbol intended for user U_i with $\mathbb{E}[|s_i|^2] = 1$, and $a_i \in (0, 1)$ is the power-allocation coefficient satisfying $\alpha_1 + \alpha_2 = 1$. Then, the received signal at the relay can be expressed as

$$y_R = h_{SR}x_S + n_R, \tag{4}$$

where $n_R \sim \mathcal{CN}(0, \sigma^2)$ denotes AWGN at the relay with variance σ^2.

After receiving the superimposed NOMA signal, the relay simultaneously performs energy harvesting and information processing through the PS-SWIPT mechanism. Consequently, the received signal in (4) serves as the input for both operations.

2.2. SWIPT mechanism and User signal reception

The relay applies PS with factor $\rho \in (0, 1)$: a fraction $(1 - \rho)$ of the received power is used for EH, and the remaining fraction ρ is used for IP. Let $\eta \in (0, 1]$ be the energy conversion efficiency. The harvested energy within the duration $T/2$ is expressed as

$$E_R = \eta(1 - \rho)P_S|h_{SR}|^2\frac{T}{2}, \tag{5}$$

and thus the relay average transmit power used for the duration $T/2$ of communication is given by

$$P_R = \frac{E_R}{T/2} = \eta(1 - \rho)P_S|h_{SR}|^2. \tag{6}$$

Simultaneously, the relay adopts AF relaying and forwards an amplified version of its received signal. Let G denote the amplification gain. In accordance with AF operation, we express G as

$$G \triangleq \sqrt{\frac{P_R}{\mathbb{E}\left[|y_{R,\mathrm{IP}}|^2\right]}}, \tag{7}$$

where $y_{R,\mathrm{IP}}$ is the IP branch signal. Under PS, the IP branch observes $\sqrt{\rho}\, y_R$, hence

$$\mathbb{E}\left[|y_{R,\mathrm{IP}}|^2\right] = \rho P_S|h_{SR}|^2 + \sigma^2, \tag{8}$$

which yields the explicit gain

$$G = \sqrt{\frac{\eta(1 - \rho)P_S|h_{SR}|^2}{\rho P_S|h_{SR}|^2 + \sigma^2}}. \tag{9}$$

Therefore, the harvested energy determines the relay transmit power through (9), while the information-processing branch determines the quality of the amplified signal through the AF gain in (9). These two quantities jointly affect the end-to-end performance observed at the users.

The relay transmit signal is $x_R = G\, y_R$. Accordingly, the received signal at user U_i, $i \in \{1, 2\}$, is

$$y_{U_i} = h_{RU_i}G\big(h_{SR}x_S + n_R\big) + n_{U_i}, \tag{10}$$

where $n_{U_i} \sim \mathcal{CN}(0, \sigma^2)$ denotes AWGN at user U_i with the same variance σ^2 as at the relay.

2.3. Signal Processing and SINR Formulation

Upon receiving y_{U_i}, each user decodes according to the NOMA principle. The decoding operations at the two users are different due to the NOMA principle. User U_1 directly detects its target signal s_1 while treating

the remaining signal s_2 as interference. In contrast, User U_2 performs SIC, which requires an additional decoding stage s_1 before recovering its own information signal s_2.

From (10) and (3), the desired signal power for s_1 at U_1 equals $P_S\alpha_1|h_{SR}|^2|h_{RU_1}|^2G^2$, while the interference from s_2 equals $P_S\alpha_2|h_{SR}|^2|h_{RU_1}|^2G^2$. The forwarded relay noise contributes $|h_{RU_1}|^2G^2\sigma^2$, and the local noise contributes σ^2. Therefore, the SINR for decoding s_1 at U_1 can be written as

$$\gamma_1 = \frac{P_S\alpha_1|h_{SR}|^2|h_{RU_1}|^2G^2}{P_S\alpha_2|h_{SR}|^2|h_{RU_1}|^2G^2 + |h_{RU_1}|^2G^2\sigma^2 + \sigma^2}. \tag{11}$$

Similarly, when U_2 decodes s_1 (prior to SIC), its received SINR can be expressed as

$$\gamma_{2\to1} = \frac{P_S\alpha_1|h_{SR}|^2|h_{RU_2}|^2G^2}{P_S\alpha_2|h_{SR}|^2|h_{RU_2}|^2G^2 + |h_{RU_2}|^2G^2\sigma^2 + \sigma^2}. \tag{12}$$

After removing s_1 via SIC, residual interference remains due to imperfect SIC. Modeling this residual term by factor δ, the SINR for decoding s_2 at U_2 is

$$\gamma_2 = \frac{P_S\alpha_2|h_{SR}|^2|h_{RU_2}|^2G^2}{\delta\, P_S\alpha_1|h_{SR}|^2|h_{RU_2}|^2G^2 + |h_{RU_2}|^2G^2\sigma^2 + \sigma^2}. \tag{13}$$

where δ is the imperfect SIC factor belong to $\delta \in [0, 1]$, where $\delta = 0$ corresponds to perfect SIC.

It should be noted that the present analysis assumes perfect CSI for signal detection and SIC processing. This assumption is commonly adopted in performance analysis studies to obtain tractable analytical expressions. In practical systems, CSI estimation errors may introduce additional decoding uncertainty and reduce the effectiveness of SIC, which would generally increase the outage probability and reduce the achievable ergodic capacity.

Tab. 2: List of Main Symbols Used in the Manuscript.

Symbol	Description
S	Source node
R	Energy-harvesting AF relay
U_1, U_2	User 1 and user 2
h_{XY}	Channel coefficient from node X to node Y
g_{XY}	Channel power gain, $\|h_{XY}\|^2$
Ω_{XY}	Average channel power gain
d_{XY}	Distance between nodes X and Y
ζ	Path-loss exponent
P_S	Source transmit power
P_R	Relay transmit power
s_i	Information symbols for U_i
α_i	NOMA power allocation coefficients
ρ	Power-splitting factor
η	Energy conversion efficiency
E_R	Harvested energy at the relay
G	AF amplification gain
σ^2	Noise variance
δ	Imperfect SIC coefficient
γ_1	SINR for decoding s_1 at U_1
$\gamma_{2\to1}$	SINR for decoding s_1 at U_2
γ_2	SINR for decoding s_2 at U_2
X	Source-relay channel power gain, $\|h_{SR}\|^2$
Y_i	Relay-user channel power gain, $\|h_{RU_i}\|^2$
Ψ	Transmit SNR, P_S/σ^2
R_i	Target transmission rate
τ_i	SINR threshold corresponding to R_i
$\phi_1(\cdot)$	Auxiliary function defined in (24)
$\phi_2(\cdot)$	Auxiliary function defined in (31)
$\phi_{\max}$	Maximum threshold used in U_2
ξ	Boundary point for EC integration of U_2
$K_1(\cdot)$	Modified Bessel-K function

3. Performance Analysis

In this section, we develop an analytical framework to evaluate the OP and the EC of the considered SWIPT-enabled AF-NOMA relay network. The analysis is consistent with the block Rayleigh fading assumption, the PS-based energy harvesting model, and the AF forwarding structure introduced in Section 2. .

3.1. Preliminaries and Definitions

For the sake of notation, we define the channel power gains as

$$X \triangleq |h_{SR}|^2, \qquad Y_i \triangleq |h_{RU_i}|^2, \tag{14}$$

where X and Y_i are independent exponential random variables with means Ω_{SR} and Ω_{RU_i}, respectively, i.e.,

$$f_X(x) = \frac{1}{\Omega_{SR}} \exp\left(-\frac{x}{\Omega_{SR}}\right), \; x \geq 0, \tag{15}$$

and

$$f_{Y_i}(y) = \frac{1}{\Omega_{RU_i}} \exp\left(-\frac{y}{\Omega_{RU_i}}\right), \; y \geq 0. \tag{16}$$

The exponential distributions directly follow from the Rayleigh fading assumption introduced in Section 2. and are extensively used in outage and ergodic-capacity analysis of wireless fading channels [27–33].

By substituting the AF gain in (9) into (11)-(13), and then normalizing all signal, interference, and noise terms by σ^2, the SINR expressions can be rewritten in terms of the normalized transmit SNR $\Psi \triangleq \frac{P_S}{\sigma^2}$.. Moreover, using $X = |h_{SR}|^2$ and $Y_i = |h_{RU_i}|^2$ allows the end-to-end channel effect to be expressed through the product XY_i. This transformation separates the desired signal, the NOMA interference, the AF-amplified relay noise, and the local receiver noise, leading to the

compact SINR representations:

$$\gamma_1 = \frac{\alpha_1 \Psi\, XY_1}{\alpha_2 \Psi\, XY_1 + \frac{\Psi}{\eta(1-\rho)} Y_1 + 1}, \tag{17}$$

$$\gamma_{2\to1} = \frac{\alpha_1 \Psi\, XY_2}{\alpha_2 \Psi\, XY_2 + \frac{\Psi}{\eta(1-\rho)} Y_2 + 1}, \tag{18}$$

$$\gamma_2 = \frac{\alpha_2 \Psi\, XY_2}{\delta\alpha_1 \Psi\, XY_2 + \frac{\Psi}{\eta(1-\rho)} Y_2 + 1}. \tag{19}$$

For $\beta > 0$ and $\gamma > 0$, the following integral identity involving the modified Bessel function of the second kind is adopted from [34] as

$$\int_0^\infty \exp\left(-\beta y - \frac{\gamma}{y}\right) dy = 2\sqrt{\frac{\gamma}{\beta}}\, K_1\left(2\sqrt{\beta\gamma}\right), \tag{20}$$

where $K_1(\cdot)$ is the modified Bessel function of the second kind of order one. The identity in (20) is a standard integral result involving the modified Bessel function of the second kind and has been widely used in wireless performance analysis over fading channels [27, 28].

3.2. OP Analysis

For the two-hop (half-duplex) transmission, we denote the target rate in (bps/Hz) by R_i and define the corresponding SINR threshold as

$$\tau_i \triangleq 2^{2R_i} - 1. \tag{21}$$

1) OP at User U_1

The outage probability at U_1 is defined as the probability that the received SINR at U_1 falls below the target data rate R_1, i.e.,

$$P_{\text{out},1} \triangleq \Pr\left(\log_2(1+\gamma_1) < R_1\right) = \Pr\left(\gamma_1 < \tau_1\right). \tag{22}$$

Using (17), the event $\gamma_1 < \tau_1$ can be rearranged as

$$X < \phi_1(\tau_1)\left(\frac{1}{\eta(1-\rho)} + \frac{1}{\Psi Y_1}\right), \tag{23}$$

where

$$\phi_1(z) \triangleq \frac{z}{\alpha_1 - \alpha_2 z}. \tag{24}$$

For $\phi_1(z)$ to be well-defined and positive, we require $\alpha_1 > \alpha_2 z$. Hence, if $\alpha_1 \le \alpha_2\tau_1$, then $P_{\text{out},1} = 1$.

Conditioning on $Y_1 = y$, the outage event in (23) becomes an upper-bound condition on X. Since X is exponentially distributed with CDF $F_X(x)$, the conditional outage probability can be directly obtained through $F_X(x)$. Averaging this conditional probability over the distribution of Y_1 yields the following expression

$$\begin{aligned} P_{\text{out},1} &= \mathbb{E}_{Y_1}\left[F_X\left(\phi_1(\tau_1)\left(\frac{1}{\eta(1-\rho)} + \frac{1}{\Psi Y_1}\right)\right)\right] \\ &= 1 - \int_0^\infty \exp\left(-\frac{\phi_1(\tau_1)}{\Omega_{SR}}\left(\frac{1}{\eta(1-\rho)} + \frac{1}{\Psi y}\right)\right) \\ &\quad \times \exp\left(-\frac{y}{\Omega_{RU_1}}\right)\frac{dy}{\Omega_{RU_1}}. \end{aligned} \tag{25}$$

After separating the constant term, (25) becomes

$$\begin{aligned} P_{\text{out},1} &= 1 - \exp\left(-\frac{\phi_1(\tau_1)}{\eta(1-\rho)\,\Omega_{SR}}\right)\frac{1}{\Omega_{RU_1}} \\ &\quad \times \int_0^\infty \exp\left(-\frac{y}{\Omega_{RU_1}} - \frac{\phi_1(\tau_1)}{\Psi\,\Omega_{SR}}\frac{1}{y}\right) dy. \end{aligned} \tag{26}$$

Applying (20) with $\beta = 1/\Omega_{RU_1}$ and $\gamma = \phi_1(\tau_1)/(\Psi\Omega_{SR})$, we obtain the closed-form OP:

$$\begin{aligned} P_{\text{out},1} &= 1 - \exp\left(-\frac{\phi_1(\tau_1)}{\eta(1-\rho)\,\Omega_{SR}}\right) 2\sqrt{\frac{\phi_1(\tau_1)}{\Psi\,\Omega_{SR}\Omega_{RU_1}}} \\ &\quad \times K_1\left(2\sqrt{\frac{\phi_1(\tau_1)}{\Psi\,\Omega_{SR}\Omega_{RU_1}}}\right). \end{aligned} \tag{27}$$

2) OP at User U_2

User U_2 is in outage if it fails to decode s_1 (for SIC) or fails to decode s_2 after SIC. Thus,

$$\begin{aligned} P_{\text{out},2} &\triangleq \Pr\left(\gamma_{2\to1} < \tau_1 \text{ or } \gamma_2 < \tau_2\right) \\ &= 1 - \Pr\left(\gamma_{2\to1} > \tau_1 \text{ or } \gamma_2 > \tau_2\right). \end{aligned} \tag{28}$$

Using (18) and (19), the events $\gamma_{2\to1} > \tau_1$ and $\gamma_2 > \tau_2$ can be rewritten as

$$X \ge \phi_1(\tau_1)\left(\frac{1}{\eta(1-\rho)} + \frac{1}{\Psi Y_2}\right), \tag{29}$$

and

$$X \ge \phi_2(\tau_2)\left(\frac{1}{\eta(1-\rho)} + \frac{1}{\Psi Y_2}\right), \tag{30}$$

where

$$\phi_2(z) \triangleq \frac{z}{\alpha_2 - \delta\alpha_1 z}. \tag{31}$$

Following, the joint success event writes

$$X \ge \phi_{\max}\left(\frac{1}{\eta(1-\rho)} + \frac{1}{\Psi Y_2}\right), \tag{32}$$

where

$$\phi_{\max} \triangleq \max\left\{\phi_1(\tau_1),\ \phi_2(\tau_2)\right\}. \tag{33}$$

If any condition in $\alpha_1 > \alpha_2\tau_1$ (for decoding s_1) and $\alpha_2 > \delta\alpha_1\tau_2$ for decoding s_2 is violated, then $P_{\text{out},2} = 1$.

Since successful decoding at U_2 requires both $\gamma_2 \to 1 \geq \tau_1$ and $\gamma_2 \geq \tau_2$, the corresponding threshold on X is determined by the more restrictive condition. Therefore, the joint decoding event can be compactly represented by the maximum threshold $\phi_{\max}$. Thus, we obtain

$$P_{\text{out},2} = \Pr\left(X < \phi_{\max}\left(\frac{1}{\eta(1-\rho)} + \frac{1}{\Psi Y_2}\right)\right), \quad (34)$$

and following the same steps as (25)-(27), we obtain

$$P_{\text{out},2} = 1 - \exp\left(-\frac{\phi_{\max}}{\eta(1-\rho)\,\Omega_{SR}}\right) \times 2\sqrt{\frac{\phi_{\max}}{\Psi\,\Omega_{SR}\Omega_{RU_2}}}\,K_1\left(2\sqrt{\frac{\phi_{\max}}{\Psi\,\Omega_{SR}\Omega_{RU_2}}}\right). \quad (35)$$

3) Insights behind Outage Analysis

The closed-form outage expressions provide several useful engineering insights. First, the outage probability is governed by both the transmit SNR and the effective channel gains, confirming that improving channel conditions or increasing transmit power enhances reliability. Second, the appearance of the factor $\eta(1-\rho)$ explicitly demonstrates the SWIPT tradeoff, where increasing the harvested-energy portion strengthens relay transmission capability but simultaneously reduces the power available for information processing. Third, the outage thresholds $\varphi_1(\tau_1)$ and $\varphi_2(\tau_2)$ reveal the direct influence of the NOMA power-allocation coefficients on decoding feasibility. In particular, inappropriate power allocation may violate the decoding conditions and lead to outage regardless of channel quality. Finally, the imperfect SIC parameter δ appears in the decoding constraint of user U_2, indicating that residual interference can become a dominant reliability-limiting factor even when the transmit SNR is high. Therefore, the derived expressions are not only useful for performance evaluation but also provide practical guidance for selecting power-allocation and power-splitting parameters in SWIPT-assisted AF-NOMA networks.

3.3. Ergodic Capacity (EC) Analysis

Due to the two-hop half-duplex relaying protocol, the instantaneous achievable rate (bps/Hz) for user U_i under the SIC-success constraint is

$$C_i = \frac{1}{2}\log_2(1+\Gamma_i). \quad (36)$$

Upon this, a convenient CDF-based identity for EC can be expressed as

$$\bar{C}_i \triangleq \mathbb{E}[C_i] = \mathbb{E}\left[\log_2(1+\Gamma_i)\right] = \frac{1}{\ln(2)}\int_0^\infty \frac{1-F_{\Gamma_i}(x)}{1+x}\,dx = \frac{1}{2\ln(2)}\int_0^\infty \frac{1-F_{\Gamma_i}(x)}{1+x}\,dx. \quad (37)$$

1) Ergodic Capacity of U_1

From (17), the CDF $F_{\gamma_1}(x) = F_{\gamma_1}(x)$ can be derived by repeating the outage steps with τ_1 replaced by a generic variable x, which yields

$$\phi_1(x) = \frac{x}{\alpha_1 - \alpha_2 x}, \qquad 0 \leq x < \frac{\alpha_1}{\alpha_2}. \quad (38)$$

Then,

$$F_{\gamma_1}(x) = 1 - \exp\left(-\frac{\phi_1(x)}{\eta(1-\rho)\,\Omega_{SR}}\right) 2\sqrt{\frac{\phi_1(x)}{\Psi\,\Omega_{SR}\Omega_{RU_1}}} \times K_1\left(2\sqrt{\frac{\phi_1(x)}{\Psi\,\Omega_{SR}\Omega_{RU_1}}}\right). \quad (39)$$

Substituting (39) into (37) gives

$$\bar{C}_1 = \int_0^{\frac{\alpha_1}{\alpha_2}} \frac{1-F_{\gamma_1}(x)}{2\ln(2)(1+x)}\,dx = \int_0^{\frac{\alpha_1}{\alpha_2}} \frac{\chi(x)}{2\ln(2)}\,dx, \quad (40)$$

where

$$\chi(x) = \frac{1}{1+x}\exp\left(-\frac{\phi_1(x)}{\eta(1-\rho)\,\Omega_{SR}}\right) 2\sqrt{\frac{\phi_1(x)}{\Psi\,\Omega_{SR}\Omega_{RU_1}}} \times K_1\left(2\sqrt{\frac{\phi_1(x)}{\Psi\,\Omega_{SR}\Omega_{RU_1}}}\right). \quad (41)$$

The integral in (40) contains both the modified Bessel function and the nonlinear function $\phi_1(x)$. As a result, obtaining a simple closed-form antiderivative is mathematically impossible. Thus, we approximate it using Gaussian-Chebyshev quadrature (GCQ) [11], i.e.,

$$\int_a^b f(x)dx = \frac{b-a}{2}\frac{\pi}{N}\sum_{n=1}^{N}\sqrt{1-\omega_n^2}\,f(x_n), \quad (42)$$

where

$$x_n = \frac{b-a}{2}\omega_n + \frac{b+a}{2}, \quad (43)$$

$$\omega_n = \cos\left(\frac{2n-1}{2N}\pi\right). \quad (44)$$

Thus, by mapping $b = \alpha_1/\alpha_2$ and $a = 0$, the EC of U_1 can be approximated as

$$\bar{C}_1 \approx \frac{1}{2\ln(2)}\frac{\alpha_1}{2\alpha_2}\frac{\pi}{N}\sum_{n=1}^{N}\sqrt{1-\omega_n^2}\,\chi\left(\frac{\alpha_1}{2\alpha_2}\omega_n + \frac{\alpha_1}{2\alpha_2}\right). \quad (45)$$

2) Ergodic Capacity of U_2

For U_2, successful decoding of s_2 requires that s_1 is decoded first. Therefore, we define an effective CDF that incorporates the SIC-success constraint:

$$F_{\gamma_2}(x) \triangleq 1 - \Pr\left(\gamma_{2\to1} \geq \tau_1,\ \gamma_2 \geq x\right), \quad x \geq 0. \quad (46)$$

Using (18) and (19), the joint success event can be expressed as

$$X \geq \phi_{\max}\left(\frac{1}{\eta(1-\rho)} + \frac{1}{\Psi Y_2}\right). \tag{47}$$

Hence, the effective CDF becomes

$$F_{\gamma_2}(x) = 1 - \exp\left(-\frac{\phi_{\max}}{\eta(1-\rho)\,\Omega_{SR}}\right) \cdot 2\sqrt{\frac{\phi_{\max}}{\Psi\,\Omega_{SR}\Omega_{RU_2}}} \times K_1\left(2\sqrt{\frac{\phi_{\max}}{\Psi\,\Omega_{SR}\Omega_{RU_2}}}\right). \tag{48}$$

It should be noted that when $\tau_1 > \alpha_1/\alpha_2$, U_2 fails to decode s_1 and thus the EC of U_2 becomes zero.

Provided that $\tau_1 < \alpha_1/\alpha_2$, the max-operator in (47) yields a natural two cases:

- **Case 1:** $\phi_1(\tau_1) \geq \phi_2(x)$, we have

$$\frac{\tau_1}{\alpha_1 - \alpha_2\tau_1} \geq \frac{x}{\alpha_2 - \delta\alpha_1 x} \Rightarrow x \leq \frac{\tau_1\alpha_2}{\alpha_1 - \alpha_2\tau_1 + \tau_1\delta\alpha_1} \triangleq \xi. \tag{49}$$

  However, $\phi_2(x)$ holds only if $x < \frac{\alpha_2}{\delta\alpha_1}$, thus the valid region of x is given by

$$x \leq \min\left\{\xi, \frac{\alpha_2}{\alpha_1\delta}\right\}. \tag{50}$$

  This means that the SIC-decoding constraint dominates, and $\phi_{\max} = \phi_1(\tau_1)$ is constant.

- **Case 2:** $\phi_1(\tau_1) < \phi_2(x)$, we have that

$$\frac{\tau_1}{\alpha_1 - \alpha_2\tau_1} < \frac{x}{\alpha_2 - \delta\alpha_1 x} \Rightarrow \xi < x < \frac{\alpha_2}{\alpha_1\delta}. \tag{51}$$

  In this case, the decoding of s_2 dominates and $\phi_{\max} = \phi_2(x)$ varies with x.

The point ξ corresponds to the intersection of the two decoding constraints, i.e., $\phi_1(\tau_1) = \phi_2(x)$. Therefore, the integration interval can be divided into two regions. In the first region, the SIC-decoding requirement dominates and $\phi_{\max}$ remains constant. In the second region, the decoding condition of s_2 becomes dominant and $\phi_{\max}$ varies with x. Thus, substituting (48) into (37) gives

$$\bar{C}_2 = \frac{1}{2\ln(2)}\int_0^{\xi}\frac{\Xi}{1+x}\,dx + \frac{1}{2\ln(2)}\int_{\xi}^{\frac{\alpha_2}{\alpha_1\delta}}\frac{\Phi(x)}{1+x}\,dx = \frac{1}{2}\Xi\log_2(1+\xi) + \frac{1}{2\ln(2)}\int_{\xi}^{\frac{\alpha_2}{\alpha_1\delta}}\frac{\Phi(x)}{1+x}\,dx, \tag{52}$$

where

$$\Xi = \exp\left(-\frac{\phi_1(\tau_1)}{\eta(1-\rho)\,\Omega_{SR}}\right) \cdot 2\sqrt{\frac{\phi_1(\tau_1)}{\Psi\,\Omega_{SR}\Omega_{RU_2}}} \times K_1\left(2\sqrt{\frac{\phi_1(\tau_1)}{\Psi\,\Omega_{SR}\Omega_{RU_2}}}\right), \tag{53}$$

$$\Phi(x) = \exp\left(-\frac{\phi_2(x)}{\eta(1-\rho)\,\Omega_{SR}}\right) \cdot 2\sqrt{\frac{\phi_2(x)}{\Psi\,\Omega_{SR}\Omega_{RU_2}}} \times K_1\left(2\sqrt{\frac{\phi_2(x)}{\Psi\,\Omega_{SR}\Omega_{RU_2}}}\right). \tag{54}$$

The remaining integral in (52) has no closed-form solution and thus, we apply the GCQ method to approximate it. Accordingly, the EC of U_2 becomes

$$\bar{C}_2 \approx \frac{1}{2}\Xi\log_2(1+\xi) + \frac{1}{2\ln(2)}\left(\frac{\alpha_2}{\alpha_1\delta} - \xi\right)\frac{1}{2}\frac{\pi}{N}\sum_{n=1}^{N} \times \sqrt{1-\omega_n^2}\,\Phi\left(\left[\frac{\alpha_2}{\alpha_1\delta} - \xi\right]\omega_n + \frac{\alpha_2}{\alpha_1\delta} + \xi\right). \tag{55}$$

4. Numerical Results and Discussions

In this section, the analytical expressions are validated through Monte-Carlo simulations and the system performance is evaluated under a normalized relay-assisted wireless setup. The Monte-Carlo simulations are used only as a numerical benchmark to verify the derived closed-form and quadrature-based expressions. Each simulated point is obtained by averaging over a sufficiently large number of independent block-Rayleigh fading realizations so that the simulation markers become stable and overlap with the analytical curves. The analytical evaluation also has low computational complexity. The outage expressions require only direct function evaluations, including the modified Bessel function, for each parameter point. For EC, the Gaussian-Chebyshev quadrature requires N quadrature nodes, leading to $O(N)$ complexity per point. In contrast, Monte-Carlo simulation requires M independent fading realizations per point, resulting in $O(M)$ complexity. Since N is much smaller than M in practical numerical evaluation, the proposed analytical method is considerably more efficient than exhaustive simulation while maintaining high accuracy. The selected parameters represent a practical baseline for cooperative NOMA/SWIPT systems. The transmit SNR $\Psi = 10$ dB corresponds to a moderate operating regime in which both noise and interference effects are visible. The target rate $R_i = 0.5$ bps/Hz is suitable for half-duplex relaying because the pre-log factor reduces the effective spectral efficiency. The power allocation $\alpha_1 = 0.6$ and $\alpha_2 = 0.4$ represents a typical

two-user NOMA setting where different power levels are assigned to support successive interference cancellation. The imperfect-SIC factor $\delta = 0.1$ models residual interference after cancellation. In addition, the energy conversion efficiency $\eta = 1$ is assumed to be constant throughout the analysis. This assumption allows the impact of SWIPT power splitting and channel fading to be isolated from the characteristics of the harvesting circuitry. In practice, the RF-to-DC conversion efficiency depends on the received power level and circuit implementation. Therefore, systems employing nonlinear energy-harvesting models may experience different quantitative performance, although the fundamental trends observed in this work are expected to remain unchanged. The normalized distances $d_{SR} = 0.5$ and $d_{RU_1} = d_{RU_2} = 1$ describe a relay-assisted topology in which the relay is located closer to the source and forwards information to both users. The pathloss exponent $\zeta = 2$ is used as a baseline free-space or low-obstruction propagation condition. Besides, it should be also noted that the presented results are obtained under ideal transceiver assumptions. In practical deployments, hardware impairments such as amplifier nonlinearity, phase noise, in-phase/quadrature imbalance, and oscillator imperfections may introduce additional distortion terms. Since these impairments reduce the effective received SINR, the outage and capacity performance reported in this work should be interpreted as an analytical benchmark for systems with sufficiently high hardware quality. For benchmarking purposes, a conventional AF-NOMA relay system employing the TSR protocol [15] is also evaluated. The benchmark uses the same channel, fading, power-allocation, and SIC parameters as the proposed PS-SWIPT scheme to ensure a fair comparison.

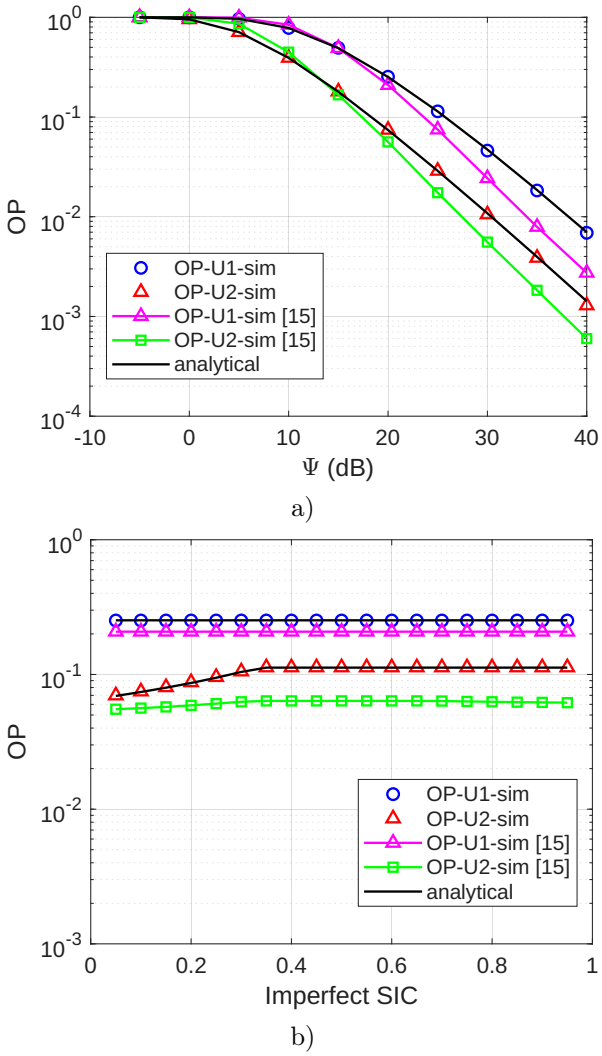


Fig. 1: OP vs SNR: a) OP versus SNR and b) OP versus imperfect SIC.

Fig. 1(a) depicts the OP as a function of the average transmit SNR, denoted by Ψ (dB). As Ψ increases, the OP of both users decreases monotonically, and this monotonic decay indicates that higher SNR improves link reliability under the considered fading and interference conditions. Moreover, the analytical curves for both users closely match with the Monte-Carlo simulation markers across the entire SNR range, and this close agreement validates the correctness of the derived outage expressions and the associated end-to-end SINR modeling. In addition, user 2 consistently achieves a lower OP than user 1 at moderate-to-high SNR values, and this performance gap suggests that the two users experience different effective interference levels and decoding constraints under power-domain superposition and successive interference cancellation. At high SNR (e.g., near $\Psi = 40$ dB), the OP of user 2 approaches the order of 10^{-3} whereas the OP of user 1 remains around the order of 10^{-2}, and this separation indicates that the reliability gain with SNR is more pronounced for user 2 than for user 1. Such a slope disparity typically arises when one user is more strongly limited by residual interference and/or compounded noise amplification effects, whereas the other user benefits more directly from increased received power. These behaviors can be physically explained from the SINR expressions in (17)-(19). As Ψ increases, the desired signal components become stronger relative to the additive noise terms, which reduces the probability that the instantaneous SINR falls below the target threshold. However, the outage improvement is not only determined by thermal noise. Since the relay employs AF forwarding, the relay noise is also amplified and forwarded to the users. Moreover, imperfect SIC leaves residual interference at user 2. Therefore, at high SNR, the system gradually becomes limited by residual interference and AF-amplified noise, which explains why the outage curves do not vanish arbitrarily fast. In addition to the proposed PS-SWIPT scheme, Fig. 1(a) also includes the TSR benchmark. It can be observed that the proposed PS-SWIPT protocol generally achieves lower outage probability than TSR in the moderate-to-high

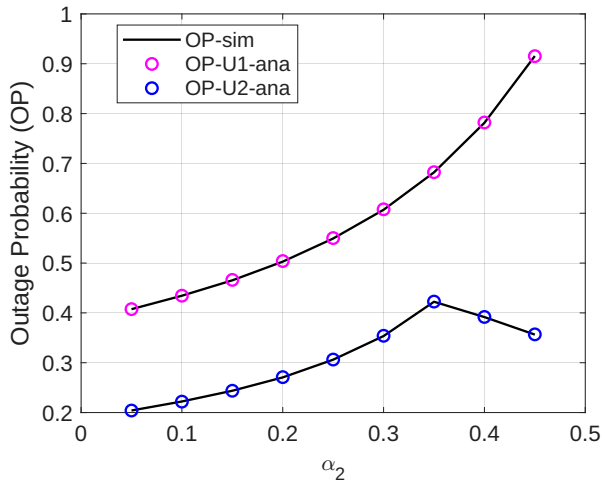


Fig. 2: OP vs α

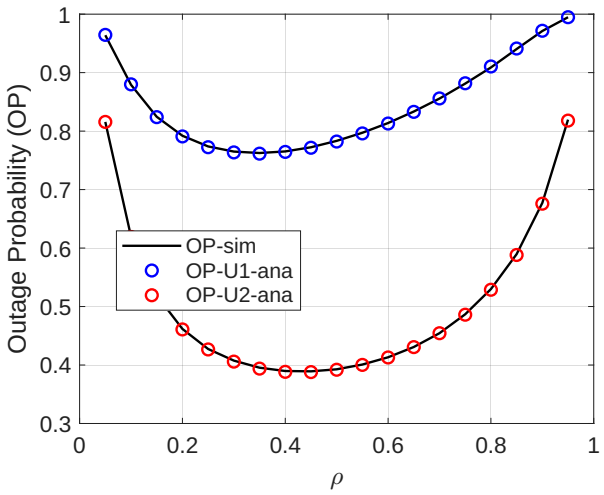


Fig. 3: OP vs ρ

SNR region. This improvement originates from the simultaneous information and energy transfer capability of PS-SWIPT, which utilizes the received RF signal more efficiently than the alternating harvesting-and-transmission operation of TSR.

Fig. 1(b) illustrates the outage probability as a function of the imperfect SIC coefficient δ. As δ increases, the residual interference remaining after SIC becomes stronger, which directly degrades the SINR of U_2 according to (13) and (19). Consequently, the outage probability of U_2 increases significantly with δ. In contrast, the performance of U_1 remains nearly unchanged because its decoding process does not involve SIC. These results confirm that the reliability of the proposed SWIPT-AF-NOMA system is highly sensitive to SIC quality and highlight the importance of efficient interference cancellation techniques in practical implementations.

Fig. 2 shows the OP versus the power-allocation coefficient α_2, and the plotted curves again include simulation results and analytical results for both users. As α_2 increases, the OP of user 1 increases steadily, and this trend follows because allocating more power to the second user necessarily reduces the available power to the first user (under normalized superposition coding), thereby degrading the effective SINR of user 1. In contrast, the OP of user 2 exhibits a non-monotonic behavior: the OP initially increases with α_2 and then decreases slightly at larger α_2 values, and this behavior indicates a coupled decoding tradeoff in which multiple decoding conditions jointly determine the outage event. Specifically, when α_2 becomes large, the power assigned to the other superimposed component becomes small, and this reduction can make the preliminary decoding step required for interference cancellation less reliable; however, further increasing α_2 strengthens the desired component after interference mitigation, which can partially restore reliability and thus reduce OP in the high-α_2 region. Consequently, Fig. 2 reveals an explicit fairness-reliability tradeoff: a small α_2 preserves user 1 reliability but can underpower user 2, whereas a large α_2 improves the power budget for user 2 but simultaneously drives user 1 toward a high outage regime. Therefore, an intermediate α_2 region becomes preferable when the design objective is to balance the outage performance between the two users rather than to optimize only one user. The above behavior follows from the fact that the NOMA power-allocation coefficients appear in both the desired-signal and interference terms of the SINR expressions. Increasing α_2 strengthens the signal intended for user 2, but it simultaneously reduces α_1 and weakens the component required for decoding s_1. Since user 2 must first decode s_1 before detecting s_2, the power allocation directly affects the SIC feasibility condition. Hence, the observed outage behavior reflects a reliability tradeoff between improving the desired signal of one user and preserving the decoding condition of the other signal. Finally, the tight match between simulation and analysis in Fig. 2 further confirms that the analytical expressions remain accurate under power-allocation sweeps, and this robustness is essential because α_2 directly governs inter-user interference and the feasibility of SIC in NOMA.

Fig. 3 depicts the OP as a function of the PS factor ρ, where $\rho \in (0, 1)$ controls how the received power at the relay is divided between energy harvesting and information processing. The convex behavior in Fig. 3 is caused by the opposite effects of ρ on information processing and energy harvesting. In the adopted PS-SWIPT architecture, the fraction ρ of the received signal is used for information processing, whereas the remaining fraction $(1 - \rho)$ is used for energy harvesting. When ρ is very small, the relay harvests more energy but the information-processing branch receives a weak signal, which degrades the quality of the am-

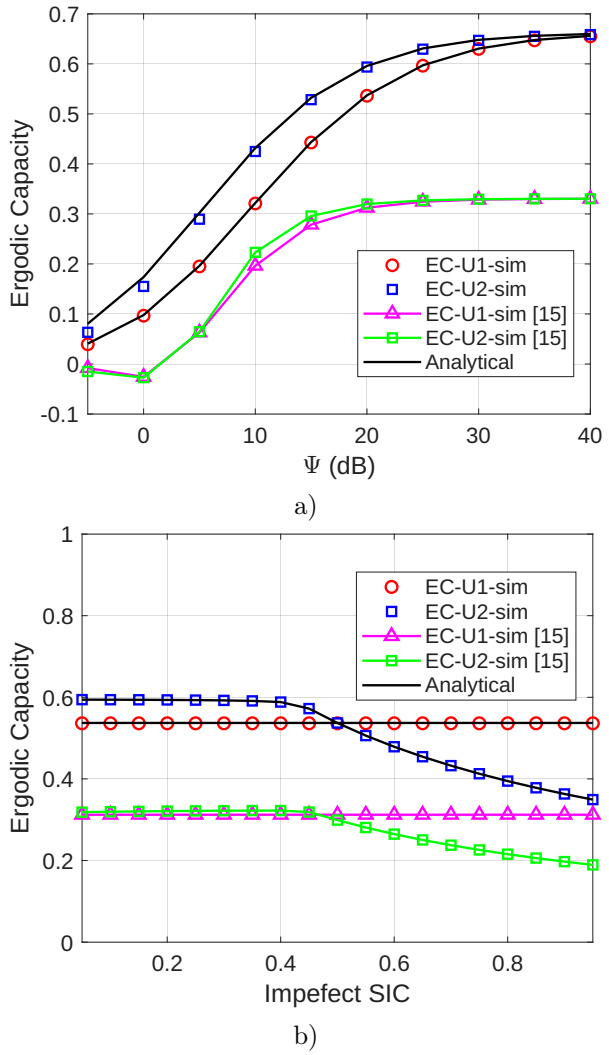


Fig. 4: EC vs SNR: a) OP versus SNR and b) OP versus imperfect SIC

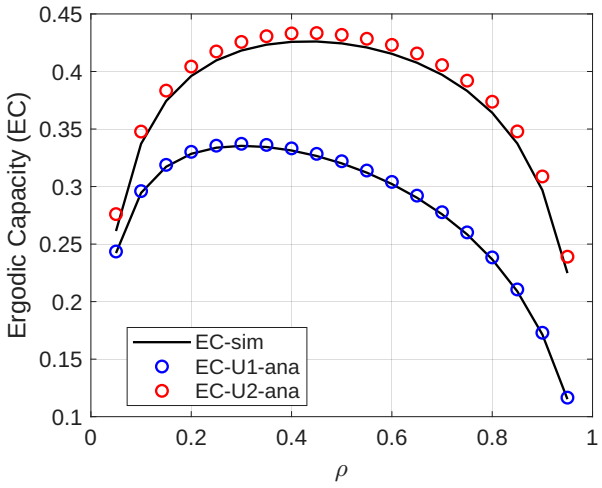


Fig. 5: EC vs ρ

plified forwarded information. When ρ is very large, the relay processes a stronger signal but harvests insufficient energy for AF forwarding. Therefore, both extreme cases degrade the end-to-end SINR, and an intermediate ρ provides the best outage performance. Moreover, user 2 maintains a lower OP than user 1 over the entire ρ range, and this persistent gap reflects the user-dependent decoding constraints and interference conditions inherent to power-domain superposition and SIC.

Fig. 4(a) presents the EC versus the average transmit SNR Ψ (dB). As Ψ increases, the EC of both users increases monotonically, and this monotonic growth indicates that higher SNR improves the long-term average achievable rate. This trend is consistent with the logarithmic relationship between achievable rate and SINR. A larger Ψ increases the average value of $\log_2(1+\gamma)$, and therefore improves the EC. Nevertheless, the EC growth becomes slower at high SNR because the system is no longer purely noise-limited. Residual interference due to imperfect SIC and the noise forwarded by the AF relay remain coupled with the useful signal, which limits the effective SINR improvement and leads to capacity saturation. The analytical EC curves closely match the simulation results across the entire SNR range, and this match validates the adopted EC evaluation method. Finally, user 2 achieves slightly higher EC than user 1 for most SNR values, and this outcome is consistent with the corresponding outage trends, indicating that the user with better reliability also attains a higher average rate under the same parameter settings. The TSR benchmark is also included in Fig. 4(a). The proposed PS-SWIPT framework consistently provides a higher ergodic capacity because information processing and energy harvesting occur simultaneously within each transmission block, whereas TSR allocates a portion of the block duration exclusively to energy harvesting. Consequently, less time remains available for information transmission in the TSR protocol.

Fig. 4(b) presents the EC under different levels of residual SIC interference. As δ increases, the average SINR available for decoding s_2 decreases, resulting in a noticeable reduction in the EC of U_2. The impact on U_1 is negligible because its decoding strategy does not depend on SIC. Therefore, imperfect SIC primarily affects the user performing interference cancellation and creates a performance gap between ideal and practical NOMA receivers.

Fig. 5 plots the EC as a function of ρ. A pronounced concave trend is observed, and this behavior indicates that an optimal intermediate ρ maximizes the long-term average rate. The concave EC behavior in Fig. 5 is also governed by the energy-information tradeoff of the PS-SWIPT relay. For small ρ, although more received power is allocated to energy harvesting through the $(1-\rho)$ branch, the signal available for information processing is weak, which limits the quality of

AF forwarding. For large ρ, the relay can process a stronger information signal, but the harvested energy becomes insufficient to support effective relay transmission. Consequently, the maximum EC is achieved at an intermediate ρ, where the relay maintains both sufficient information-processing quality and sufficient harvested energy for forwarding. Therefore, the peak in Fig. 5 captures the fundamental SWIPT tradeoff between "*harvesting more energy for forwarding*" and "*preserving sufficient signal power for reliable IP*". In addition, the user-dependent curves remain separated over the entire ρ range, and the upper curve consistently provides higher EC than the lower curve. This persistent gap implies that the two users experience different effective interference conditions and decoding requirements under power-domain superposition and SIC. Finally, the simulation markers closely track the analytical curves, and this close match confirms that the analytical EC approximation captures the impact of ρ accurately and remains reliable for optimizing the PS design.

5. Conclusion

An EH AF relay assisted NOMA framework has been analyzed to quantify OP and EC under block Rayleigh fading, path loss, and imperfect SIC. Analytical expressions for OP and tractable EC evaluation were established and verified through Monte-Carlo simulations, thereby demonstrating high accuracy across SNR and parameter variations. The numerical observations indicate that increasing transmit SNR improves reliability, whereas power allocation introduces an inherent user-dependent outage gap that must be managed to achieve fairness. The PS factor at the relay plays a decisive role because it governs a fundamental tradeoff between harvested energy for forwarding and retained power for IP. Consequently, both OP and EC are optimized at an intermediate PS region rather than at extreme values. Overall, the presented results highlight that careful joint selection of power allocation and PS is essential for balancing reliability and throughput in wireless-powered cooperative NOMA, particularly when residual interference due to imperfect SIC cannot be neglected.

Future work may extend the current framework by incorporating imperfect CSI, practical transceiver hardware impairments, and nonlinear energy-harvesting models to further bridge the gap between analytical evaluation and real-world deployment conditions.

Author Contributions

Quang-Sang Nguyen and Quang-Sy Vu proposed the main ideas, mathematical derivations, and perform Monte Carlo sim- ulations. Dinh-Tung Vo and Do Hoang Minh Han prepared the writing draft. Finally, Duc Ngoc Minh Dang and Rupak Kharel supervised the overall work, and revised the final version of the manuscript. All authors reviewed and approved the final manuscript.

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